

Fire Extinguisher Analysis on Luther College Campus Buildings

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Introduction

- Fire extinguishers should be located throughout each building and be readily accessible.
- From 2011-2015, U.S. fire departments responded to an estimated annual average of 4,100 structure fires, causing an average of 35 civilian injuries and \$14 million in direct property damage (NFPA).
- OSHA requires access to a fire extinguisher within a 75-foot travel-distance.
- The goal of this study is to analyze fire extinguisher counts by building and the density of fire extinguishers based on building area to discover any trends.

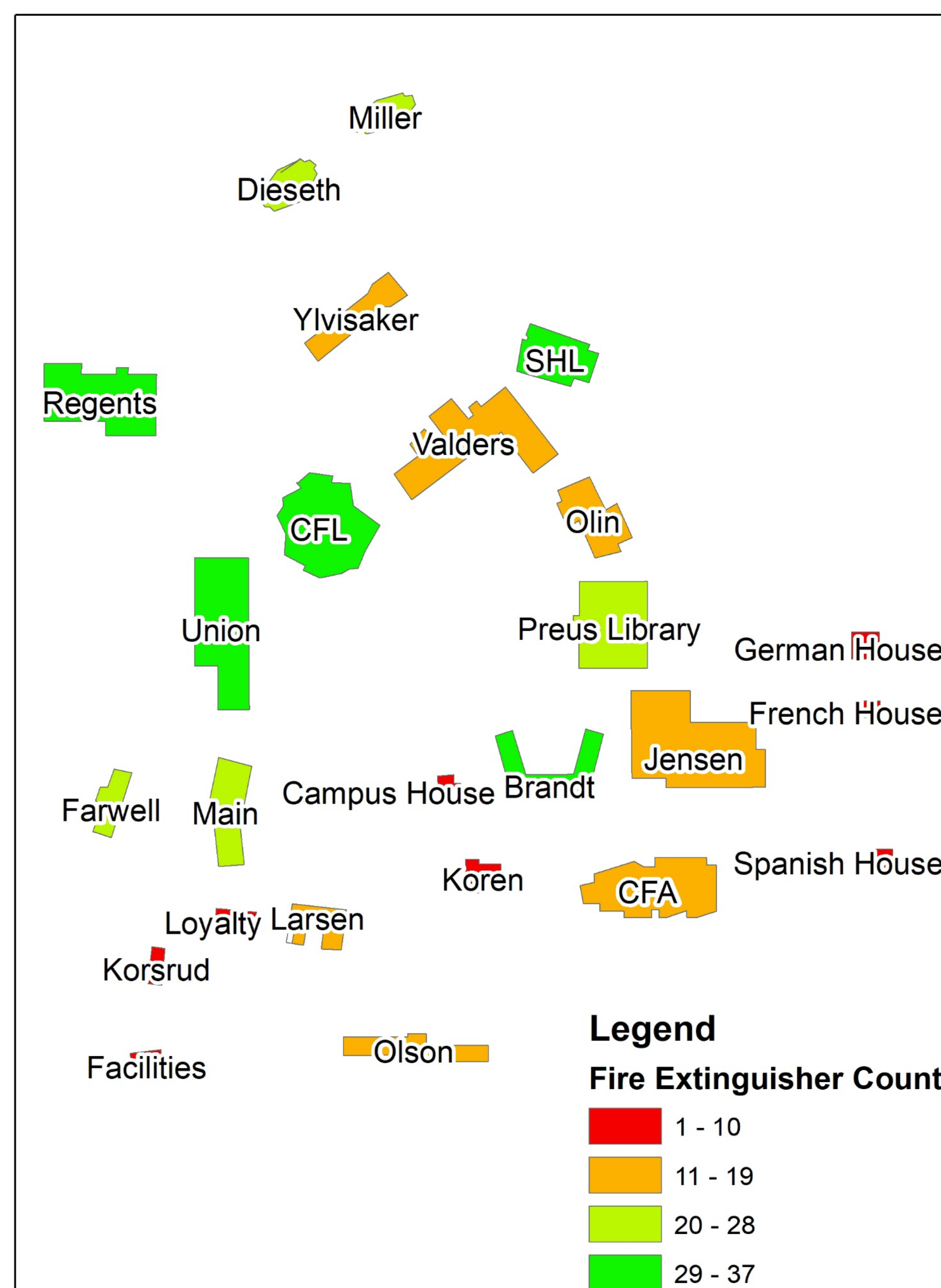


Figure 1. Red-to-green scaled choropleth map of total fire extinguisher count per building.

Methods

- Obtained data from facilities services of written fire extinguisher locations on Luther's campus.
- Hand plotted relative fire extinguisher locations on printed floor plans.
- Digitized polygons and points using ArcMap.
- The examined variables included the total count of fire extinguishers and the estimated total area of each building.

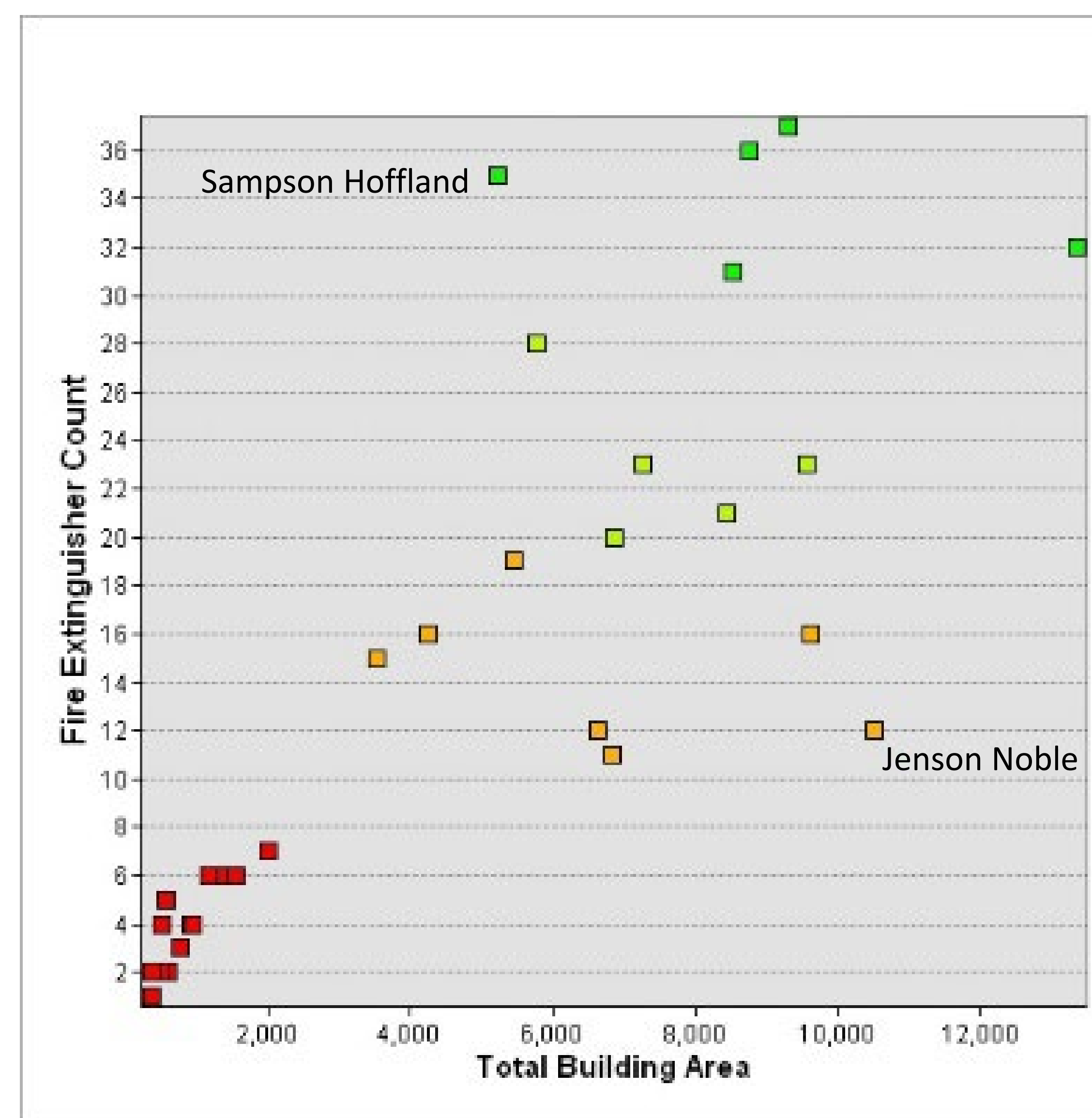


Figure 2. Scatter plot fire extinguisher count against total building area, colored by fire extinguisher count.

References

"Campus and Dorm Fires." *NFPA Reports - Fires in the United States*, National Fire Protection Association, www.nfpa.org/public-education/by-topic/property-type-and-vehicles/campus-and-dorm-fires.

"Extinguisher Placement and Spacing." *Occupational Safety and Health Administration*, www.osha.gov/SLTC/etools/evacuation/portable_placement.html.

Fire extinguisher location packet courtesy of Facilities Services

Results/Discussion

- There is a correlation between the estimated total area and the total fire extinguisher count of each building.
- We noticed two particular buildings of interest: Sampson Hoffland and Jensen Noble.
- Since Sampson Hoffland houses the physical science laboratories, we expected to observe a high fire extinguisher count for its low area.
- Jensen Noble had the least amount of fire extinguishers for its size.
- After creating the 75-foot buffer zone around the fire extinguishers in Jensen Noble, we noticed areas slightly outside of the buffered zone.
- The amount of fire extinguishers in each building does not completely rely on the building's size, but rather, the building's purpose.

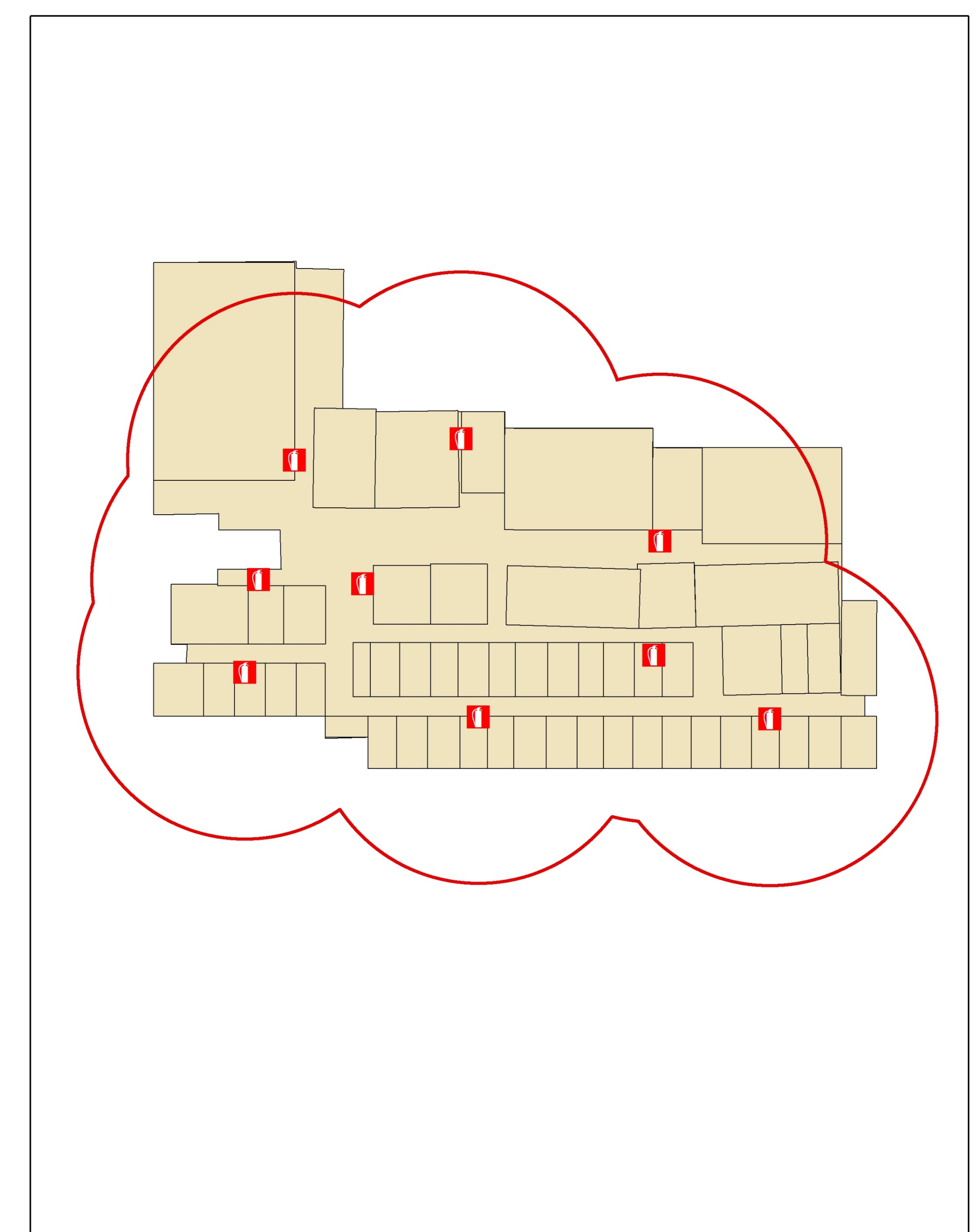


Figure 3. Digitized polygon of Jensen Noble with fire extinguisher locations and 75-foot buffers to mirror OSHA fire safety standards.